Execution methodology

DEMONSTRATION EXPORT - NOT SUBMISSION READY - NOT APPROVED FOR ANY REAL TENDER | approvals by: test principals in environment 'test'

SYNTHETIC TEST DATA - FICTIONAL CUSTOMER - NOT A REAL TENDER - NOT A MARKET QUOTE - NOT FOR SUBMISSION

Project ID	PRJ-9101
Project name	Dammam Industrial Utilities - Synthetic Tender Closure
Customer	Gulf Process Utilities Co. (FICTIONAL SYNTHETIC CUSTOMER)
Document ID	PRJ-9101-D18
Title	Execution methodology
Revision	2
Issue date	2027-02-15
Status	Preliminary (DEMONSTRATION)
Author	test-estimator
Reviewer	not yet reviewed
Estimate version	v2
Estimate fingerprint	48d0f9242dc910b7
Template	T-METHODOLOGY@1.0.0
Confidentiality	CONFIDENTIAL (synthetic test data) - no real customer or supplier data

Execution and constructability review

Key	topic	content	Status
wbs	Wbs	Work breakdown: 23 activities grouped by BOQ trade (civil, structural, piping, mechanical, electrical, ELV, commissioning).	Draft
sequence	Sequence	Execution sequence follows the programme dependencies: mobilisation, excavation, pipe-rack foundations, steel, support fabrication, piping, welding, NDT, hydrotest, backfill/coating, E&I, commissioning, demobilisation.	Draft
work_fronts	Work Fronts	Work fronts per activity are limited to the values entered on the activities (maximum 3).	Draft
plant	Plant	One mobile crane serves steel erection and pump-skid setting; a telehandler serves pipe installation (hire terms in the plant exposure report).	Draft
workforce	Workforce	Selected profile Mid-sized crew (SYNTHETIC): peak direct manpower 18.	Draft

Key	topic	content	Status
procurement	Procurement	Pump skids are a long-lead item; procurement-constrained dates are shown in the programme.	Draft
inspection	Inspection	Hold points follow the ITP register; NDT sample size is subject to the compliance deviation D-1.	Draft
interfaces	Interfaces	The fire-water tie-in is an interface with the existing ring main owned by the customer.	Draft
risks	Risks	Programme risks: hard-material excavation, pump skid delay, tie-in isolation windows (see the risk register).	Draft
temporary_works	Temporary Works	Temporary works: trench shoring (BOQ C-012) and scaffolding access hire (BOQ C-052); design of temporary works is by a competent engineer (not generated here).	Draft
access	Access	Access and working windows follow the project calendar (shutdown and reduced-hour windows are shown in D19); the customer's site access rules are a recorded required input.	Draft
testing_commissioning	Testing Commissioning	Hydrotest and flushing are a fixed 8-day activity, commissioning a fixed 10-day activity after coating, cable and fibre completion.	Draft
hse_interfaces	Hse Interfaces	HSE interfaces: hot-work and excavation permits (permit register), tie-in isolation with the existing ring main (interface register).	Draft
constraints	Constraints	Constraints and exclusions: see D09; unverified norms and the synthetic basis are listed in D37.	Draft

Execution sequence (work activities)

Activity	Name	Predecessors	Crews	Work fronts	Norm
A-MOB	Mobilisation		1	1	
A-EXC	Trench excavation	A-MOB	2	2	N-EXC
A-EXCH	Hard-material excavation	A-EXC	1	1	N-EXCH
A-SHORE	Trench shoring	A-MOB	1	1	N-SHORE
A-FORM	Pipe-rack formwork	A-EXC	2	2	N-FORM
A-REBAR	Pipe-rack reinforcement	A-EXC	2	2	N-REBAR
A-POUR	Pipe-rack concrete	A-FORM,A-REBAR	1	1	N-POUR
A-STEEL	Pipe-rack steel erection	A-POUR	2	2	N-STEEL
A-SUPP	Pipe support fabrication	A-POUR	1	1	N-SUPP
A-PUMP	Pump skid setting	A-POUR	1	1	N-PUMP

Activity	Name	Predecessors	Crews	Work fronts	Norm
A-PIPE100	Pipe installation DN100	A-STEEL,A-SUPP	3	3	N-PIPE
A-PIPE50	Pipe installation DN50	A-STEEL,A-SUPP	2	2	N-PIPE
A-WELD	Butt welding	A-PIPE100,A-PIPE50	3	3	N-WELD
A-NDT	Weld radiography	A-WELD	1	1	N-NDT
A-HYDRO	Hydrotest and flushing	A-NDT,A-PUMP	1	1	
A-BACK	Backfill	A-HYDRO	1	1	N-BACK
A-ROAD	Road reinstatement	A-BACK	1	1	N-ROAD
A-COAT	Coating	A-HYDRO	1	1	N-COAT
A-TRAY	Cable tray	A-STEEL	1	1	N-TRAY
A-CABLE	Cable pulling	A-TRAY	1	1	N-CABLE
A-ELV	Fibre backbone	A-TRAY	1	1	N-FIBRE
A-COMM	Testing and commissioning	A-COAT,A-CABLE,A-ELV,A-ROAD	1	1	
A-DEMOB	Demobilisation	A-COMM	1	1	

Assumptions, exclusions and limitations

- Quantities are human-measured BOQ quantities; the application does not measure drawings.
- Figures are taken from frozen estimate version v2 (fingerprint 48d0f9242dc910b7).
- SYNTHETIC / DEMONSTRATION data: rates, quotations, productivity norms, customer and responses are fictional test inputs, not market evidence.
- Methodology topics are entered and reviewed by a competent person; none is generated as engineering advice.

Source references

- Frozen estimate version v2 (fingerprint 48d0f9242dc910b7), totals and pricing schedule
- Registers in the application database (see document D33 for received documents, D08 for assumptions, D25 for queries)